Generative AI and Human Knowledge Sharing: Evidence from A Natural Experiment [*]

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Abstract

Generative AI, known for its content creation capabilities, possesses great potential to reshape knowledge-sharing activities. By leveraging a unique policy of a leading online Q&A platform that introduces generative AI answers, we explore the effects of generative AI answers on human contributor participation in knowledge-sharing platforms. Our findings demonstrate the positive effects of the generative AI answers on human contributor participation, both in terms of quantity and length of the subsequent answers to a question. We propose two potential mechanisms: the AI-conformity effect, where users contribute more by learning from generative AI as a credible information source, and the AI-differentiation effect, where users are motivated to provide distinct human perspectives that diverge from AI. Notably, a more nuanced investigation into human-AI answer similarity indicates that generative AI answers prompt users to provide answers more aligned with the AI-generated answers, consistent with the AI-conformity effect rather than the AI-differentiation effect. Our findings reveal that the influence of generative AI answers is more pronounced for objective and under-answered questions, as well as among non-expert contributors compared to expert contributors. In a follow-up randomized experiment, we offer corroborative evidence for the results of empirical analysis and directly test the mechanism. The experimental results indicate that generative AI answers increase participants' conformity behavior, thereby promoting greater participation, further supporting the AI-conformity effect. Our research highlights the potential of integrating generative AI to motivate human participation in knowledge generation and dissemination, adding to the burgeoning body of work on how generative AI influences human knowledge sharing.

Keywords: generative AI, knowledge sharing, Q&A platform, AI-human interaction, natural experiment, randomized experiment

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1. INTRODUCTION

In this digital era, online knowledge-sharing communities such as question-and-answer (Q&A) platforms play a central role in facilitating the creation, dissemination, and exchange of knowledge. The vitality of these platforms heavily relies on the collective wisdom and voluntary knowledge contribution from their users (Cao et al., 2024; Qiu & Kumar, 2017). However, these communities often face challenges in attracting and retaining active contributors (Kuang et al., 2019). Recent advancements in generative Artificial Intelligence (AI) have greatly improved its ability to provide answers (Edelman & Abraham, 2023). ChatGPT, for example, achieved an impressive milestone of 100 million monthly active users in just two months after its launch on November 30, 2022, making it the fastest-growing consumer application in history (Hu, 2023). This greatly raises the possibility that knowledge-sharing communities could leverage generative AI to supplement knowledge contributions on their platforms (Chui et al., 2023). Nonetheless, concerns persist regarding the incorporation of the content generated by AI (Susarla et al., 2023). While generative AI could potentially enhance knowledge acquisition, there is a need to carefully assess the benefits and costs associated with incorporating generative AI and its potential impact on knowledge-sharing dynamics.

1.1 Motivations

In recent years, the field of generative AI technology has made remarkable advancements, bringing about techniques such as ChatGPT, Bard, and Midjourney (Susarla et al., 2023). These techniques have rapidly gained immense popularity. Built on large language models and massive databases, these generative AI models exhibit impressive ability for content creation (Burtch et al., 2024; OpenAI, 2023a). The extensive knowledge base and 24/7 availability have made generative AI models valuable knowledge contributors on the Internet.

The growing proficiency of generative AI in replying to knowledge requirements attracts attention from existing knowledge-sharing communities. Yet, interestingly, these communities exhibit distinct attitudes when it comes to embracing generative AI. For instance, StackOverflow and Quora, two prominent Q&A platforms, exhibit starkly different approaches. While StackOverflow chooses to ban the use of ChatGPT and similar technologies, citing concerns about potential accuracy issues (Makyen, 2022), Quora, in contrast, proactively integrates generative AI and even showcases AI-generated answers in a prominent position on question pages as a novel IT-based feature, believing that some questions "can possibly benefit from a different perspective" (Qureshi, 2023). This discrepancy in platform policies reflects an ongoing debate about the role and impact of AI-generated content in knowledge-sharing activities (Lysyakov & Viswanathan, 2023; Susarla et al., 2023). Despite the practical and theoretical relevance of understanding how generative AI impacts human knowledge contribution in online knowledge-sharing communities, the scientific investigations in this area remain far from comprehensive. Hence, this research endeavors to fill this gap and contribute to this emerging area of research.

1.2 Research Questions and Contribution

The advent of several large-scale generative AI models in recent times enables generative AI to potentially shape interactions within online knowledge-sharing platforms such as Q&A platforms. This surge in generative AI development has spurred a growing interest among researchers, leading to an exploration of how generative AI influences knowledge-sharing dynamics on Q&A platforms (e.g., Burtch et al., 2024; Sanatizadeh et al., 2023). Existing research in this branch has primarily been carried out within contexts where generative AI functions as an external source of knowledge provision and tools for answer and question generation for platform users (Shan & Qiu, 2023; Xue et al., 2023). It is worth noting that our study focuses on a more direct and visible manner of incorporating generative AI, wherein the platform presents answers from generative AI directly on the question page. Under this circumstance, the impact of generative AI on platform dynamics remains unexplored. On the one hand, the incorporation of generative AI can discourage knowledge contribution on Q&A platforms. For instance, when human knowledge contributors notice generative AI answers, they might conclude that the Q&A community's demand for answers is already fulfilled by the generative AI, given its ability to readily provide information to those seeking knowledge (Sanatizadeh et al., 2023). Consequently, their perceptions of social interactions and reciprocations within the knowledge-sharing community may be impeded. This could result in decreased motivation among human knowledge contributors to provide their own knowledge to the communities (Jin et al., 2015; Mustafa & Zhang, 2022).

On the other hand, however, introducing generative AI answers can bring about greater knowledge contribution to knowledge-sharing platforms. In various domains, research has shown the capability of AI to facilitate human content creation (e.g., Karimi et al., 2020; Pini et al., 2019). With its extensive capacity for generating, presenting, and restructuring content (Susarla et al., 2023), generative AI has the potential to offer credible information and perspectives as reference for potential contributors, thereby lowering uncertainty about how to respond to questions in a knowledge-sharing context (Shan & Qiu, 2023). For instance, when answering questions on complex topics such as climate change, human contributors may hesitate to contribute because they feel that their personal knowledge is insufficient to provide an accurate and satisfactory answer. Generative AI, however, based on large language models trained on vast corpora of scientific literature and general knowledge, can analyze these topics and offer comprehensive perspectives. By providing relevant information and demonstrating what constitutes an appropriate contribution, the generative AI answer may help reduce uncertainty and make the act of contributing feel easier (Deutsch & Gerard, 1955). Therefore, we put forward our first research question: (1) *What is the impact of the introduction of* *generative AI answers on human contributor participation?*

We investigate this question using data obtained from a leading online Q&A platform, Quora. In January 2023, Quora launched a policy introducing generative AI as the knowledge provider and prominently displaying AI-generated answers on the question pages of some, but not all, questions. By taking advantage of this feature of the policy, we employ a difference-in-differences (DID) analysis to examine how the introduction of generative AI answers impacts human knowledge-contributing participation. Our analysis reveals a significant increase in human contributor participation, both in terms of answer quantity and length, following the introduction of generative AI answers. Given the ongoing debate and the varying stances regarding the adoption of AI in the knowledge-sharing industry, our findings regarding the first research question carry significant practical importance. The positive effects on contributor participation revealed by our empirical analysis emphasize the benefits of incorporating generative AI answers. This could offer valuable guidance for knowledge-sharing platforms to shape their policies when considering the integration of generative AI.

Nevertheless, the mechanism underlying the effects remains unclear. To gain deeper insights into the impact of generative AI answers, we undertake a detailed investigation into the content of subsequent human answers. Specifically, our focus is on understanding how the content of human answers after the policy launch is related to that provided by the generative AI. Based on the existing research on the human-AI interaction (Jussupow et al., 2021; Luo et al., 2019), we propose that two distinct mechanisms could potentially explain the observed effects: the AI-conformity effect and the AI-differentiation effect. The AI-conformity effect suggests that human users contribute more because the generative AI answer serves as a clear and credible source for reference. By reducing users' uncertainty about how to contribute, generative AI answers make it easier for them to respond and often prompt users to align their answers with the content or framing of the generative AI answer (Salomons et al., 2018).

Conversely, the AI-differentiation effect posits that users contribute more because they are motivated to assert unique value or showcase expertise by intentionally providing distinct human perspectives that diverge from those of generative AI. The inclination to differentiate from AI is supported by research showing that individuals diverge from AI outputs in tasks that express personal competence, such as knowledge-sharing activities, to preserve a psychological commitment to human uniqueness (De Freitas et al., 2023; Stein et al., 2019). The AI-conformity effect will be supported if there is an increase in the similarity between user answers and generative AI answers. In contrast, the AI-differentiation effect will be supported if the similarity between user and generative AI answers decreases. Accordingly, we introduce our second research question: (2) *How does the introduction of generative AI answers affect the content of subsequent human answers?*

Our findings indicate that the introduction of generative AI answers increases subsequent human-AI answer similarity, supporting the mechanism of the AI-conformity effect. In other words, human users are more likely to enhance their contribution by learning from information provided in AI-generated responses and aligning their answers with those of AI. These findings suggest that while the introduction of generative AI answers stimulates greater human contributor participation, it also leads to increased content similarity between AI and human answers (potentially reducing the diversity of human answers), which is an important aspect that platform operators and managers should take into account when adopting and integrating AI technologies into knowledge-sharing communities.

Previous literature suggests that the impact of AI varies with the nature of the task or context, especially in how people perceive and use AI-generated content. For example, compared to humans, AI is considered more proficient and credible in the utilitarian realm than in the hedonic realm (Longoni & Cian, 2022). Building on this reasoning and considering the specific context of knowledge sharing, we pose our third research question: (3) *Does the* *introduction of generative AI answers have heterogeneous effects on different types of questions?* To investigate the question, we introduce two moderators−question's objectivity and answer inadequacy−for their relevance in our research context. The first moderator, question objectivity, captures whether a question is objective or subjective. As indicated by prior literature (Castelo et al., 2019), people tend to perceive AI-generated content as more reliable when applied to objective tasks. As a result, when human contributors refer to AI-generated answers, they may be more inclined to rely on such content for objective questions compared to subjective ones, viewing it as a credible source of information. Hence, the impact of generative AI answers may differ depending on the objectivity of the question. The second moderator, answer adequacy, reflects the volume of existing human answers associated with a given question. Questions with fewer existing answers, referred to as under-answered questions, typically involve higher uncertainty about what constitutes a credible or appropriate contribution. This uncertainty may make such questions more challenging for humans to address and, in turn, may increase their reliance on informational cues provided by AI-generated answers (Thomas-Hunt et al., 2003). Accordingly, answer adequacy may influence the extent to which generative AI answers affect human contributor participation.

Our results reveal that generative AI answers result in significantly larger increases in both the quantity and length of human answers for objective questions compared to subjective ones, as well as for under-answered questions compared to well-answered ones. These findings reinforce the AI-conformity effect by suggesting that human contributors are more inclined to treat generative AI answers as credible informational resources when answering objective or under-answered questions. Importantly, additional analysis using human-AI answer similarity as the dependent variable also reveals greater positive effects of introducing generative AI answers for objective (vs. subjective) questions and under-answered (vs. well-answered) questions, offering direct evidence that contributors conform more to AI-generated content under these conditions. These findings regarding the third research question provide valuable practical guidance for knowledge-sharing communities to better leverage the recognized capabilities of generative AI in content creation, reminding them of the importance of contextual factors, such as task objectivity and content scarcity, in moderating the influence of generative AI on user behavior.

Beyond the influence of task contexts, contributors with different characteristics may respond differently to the introduction of generative AI answers. In knowledge-sharing activities, both non-experts and experts play essential but distinct roles. Platforms often rely on a small number of experts to maintain content quality and credibility, but they also benefit from broad participation by non-experts to keep discussions active and diverse. Given these distinct contributions, it is important to examine how contributors with varying levels of expertise respond to the introduction of generative AI answers. This leads us to our fourth research question: (4) *How does the introduction of generative AI answers affect the participation of contributors with varying levels of expertise?* We find that the introduction of generative AI answers increases participation from both expert and non-expert contributors, with a disproportionately higher increase in contributions from non-expert contributors. Compared to experts, non-experts often lack extensive domain knowledge and may struggle to access high-quality information independently (Markus et al., 2002). AI-generated answers offer accessible and credible informational cues, which are especially valuable for non-experts facing greater uncertainty about how to contribute. As a result, non-experts are more influenced by the presence of AI-generated answers.

Furthermore, we conduct a randomized experiment to provide stronger causal evidence for our findings. One concern regarding the causal identification is the potential violation of the Stable Unit Treatment Value Assumption (SUTVA) in our natural experiment. Specifically, although our empirical analysis is conducted at the question level, contributors on the Q&A

platform may be exposed to both questions that include AI-generated answers and those that do not. To mitigate this concern, our experiment randomizes participants at the individual level, ensuring that each participant is exposed to a single condition. The experimental results validate the positive effects of generative AI answers on human contributor participation. Moreover, by directly measuring conformity behavior, we provide evidence supporting the AI-conformity effect as an underlying explanation for the observed increase in participation. The findings from the randomized experiment further support the results of the natural experiment and provide additional evidence for the AI-conformity effect as an underlying explanation for the observed increase in user participation following the introduction of generative AI answers.

This study holds valuable implications for both theoretical establishment and practical applications. Theoretically, our work enriches knowledge-sharing literature, especially that involving the incorporation of AI. Amidst a growing number of research delving into the impact of generative AI on knowledge-sharing activities, our work significantly contributes to this research stream in the approach and depth of our investigation. First, by focusing on the context of embedded AI tools, our work complements previous studies that predominantly rely on an external policy—the launch of ChatGPT—as the focal treatment (e.g., Burtch et al., 2024; Li & Kim, 2024; Sanatizadeh et al., 2023; Xue et al., 2023). We demonstrate that the platform's introduction of generative AI answers encourages subsequent human contributor participation. These results differ from those of previous research, which shows external generative AI tools as substitutes for human knowledge supply that reduce platform engagement (Sanatizadeh et al., 2023; Xue et al., 2023). Our research, therefore, illuminates how different methods of implementing generative AI (i.e., external vs. embedded) can lead to distinct outcomes, enriching our understanding of AI's role in shaping user behavior.

This research can also provide valuable insights for practitioners. The varying attitudes among knowledge-sharing communities regarding the integration of generative AI answers reflect an ongoing debate concerning the advantages and drawbacks of such integration (Makyen, 2022; Qureshi, 2023). Our findings can serve as a guide for managerial strategies aiming at utilizing the content-creation capabilities of generative AI. Our research shows that introducing generative AI answers in knowledge-sharing activities can increase contributor participation in knowledge contribution. However, it may lead to a loss of original perspectives. Additionally, we identify the moderating roles of question objectivity and answer inadequacy in the observed effects on contributor participation and also explore the nuances in the responses of expert and non-expert contributors. These findings offer valuable guidance for practitioners in the knowledge-sharing field on how to effectively leverage generative AI.

2. LITERATURE REVIEW

In this section, we provide an overview of existing research and theories that are closely related to our research. More specifically, we review theoretical and empirical findings regarding generative AI in knowledge sharing and literature on human conformity to and differentiation from non-human agents.

2.1 Generative AI in Knowledge Sharing

Generative AI typically refers to a category of AI models designed to produce novel content, including text, images, or other forms of media (Susarla et al., 2023). Among generative AI models, large language models are a specific subset trained to facilitate user interaction through natural language (OpenAI, 2023a). In the context of our research, which focuses on knowledge-sharing activities, these large language models hold particular relevance. Particularly following the introduction of ChatGPT by OpenAI in November 2022, the remarkable content-creation capabilities of large language models have drawn substantial attention from researchers in the field of information systems (Dwivedi et al., 2023; Susarla et al., 2023).

An expanding body of literature begins to investigate the effects of generative AI on subsequent knowledge-sharing activities. The majority of research (i.e., Burtch et al., 2024; Li

& Kim, 2024; Sanatizadeh et al., 2023; Xue et al., 2023) indicates that the availability of generative AI tools, such as ChatGPT, typically decreases the volume of questions and/or answers, as these tools act as substitutes for human-based knowledge. For example, Burtch et al. (2024) demonstrate that the introduction of ChatGPT reduces visits to StackOverflow due to fewer user-generated questions, which subsequently leads to fewer answers. Sanatizadeh et al. (2023) also observe that the introduction of ChatGPT reduces both the number of contributors and the amount of content. Previous literature typically considers both the demand side (questions asked) and the supply side (answers provided), acknowledging that these aspects dynamically influence each other (Sanatizadeh et al., 2023; Wang & Zheng, 2024). For instance, Wang and Zheng (2024)'s work suggests that banning the use of ChatGPT leads to a decline in the quality of answers, which negatively affects the quality of subsequent questions in a dynamic interplay.

Our research complements prior studies by focusing on embedded AI tools (generative AI systems that are integrated directly into existing platforms, rather than functioning as standalone tools), which operate differently from external tools like ChatGPT. This focus offers new insights into how the mode of implementing generative AI (i.e., external vs. embedded) can lead to different outcomes, providing a complementary understanding of AI's role in influencing user behavior. Moreover, in our study, AI-generated answers are displayed on question pages only after the question has been posted, without directly affecting users' question-asking activities. This setup allows us to examine the impact of AI-generated answers on users' knowledge contribution behavior independently of changes in knowledge demand. Additionally, it enables us to explore how users respond when AI-generated answers are passively presented to them, without actively seeking them out as in previous research (e.g., Burtch et al., 2024; Sanatizadeh et al., 2023).

Previous research has also delved into quality changes in user-generated content (Liu et al., 2020). For example, Wang and Zheng (2024) find Stack Exchange's ban on creating answers with generative AI leads to quality declines in both questions and answers. While previous studies (e.g., Wang & Zheng, 2024; Zhang et al., 2024) have primarily focused on the "standalone" quality of user-generated content, our research investigates a complementary aspect: the relationship between subsequent user-generated content and AI-generated content when generative AI is integrated into the platform. We compare and contrast our present research to existing literature to illustrate the foundation for our work's distinctive contributions and unique place within the field (for a thorough comparison, please refer to Appendix A).

2.2 Human Conformity to and Differentiation from Non-human Agents

As indicated by findings in social science research, an individual's needs to conform to and to differentiate from others frequently coexist (Brewer, 1991; Brewer, 1993). Social conformity refers to the process through which individuals adjust their perceptions, attitudes, and behaviors to align with others within the social contexts (Sun et al., 2019). Conformity can involve aligning with others to meet norms or standards, as well as aligning based on perceived knowledge or correctness (Cialdini & Goldstein, 2004; Deutsch & Gerard, 1955). In contrast, social differentiation represents the process through which individuals are inclined to highlight their distinctiveness by diverging from others (Lysyakov & Viswanathan, 2023).

A widely adopted paradigm of exploring human responses to computers or AI technologies is "computers are social actors," also referred to as "CASA," which denotes that individuals treat technologies with human features as social actors and apply rules originated from human relationships (Nass et al., 1994; Schanke et al., 2021). Indeed, social conformity and differentiation, originally observed among people, are observed in a much broader scope in this technologically advanced era. Individuals are found to display a tendency for conformity and differentiation when they interact with non-human agents such as robots and artificial intelligence systems (Lysyakov & Viswanathan, 2023; Salomons et al., 2021; Vollmer et al., 2018). For example, people are found to conform their answers to those of the robot when responding to simple cognitive tasks together with a robot (Qin et al., 2022). In a crowdsourcing context, after the crowdsourcing platform introduces an AI system capable of generating logos, high-capability designers opt for distinctive designs, differentiating them from those designed by the AI system (Lysyakov & Viswanathan, 2023).

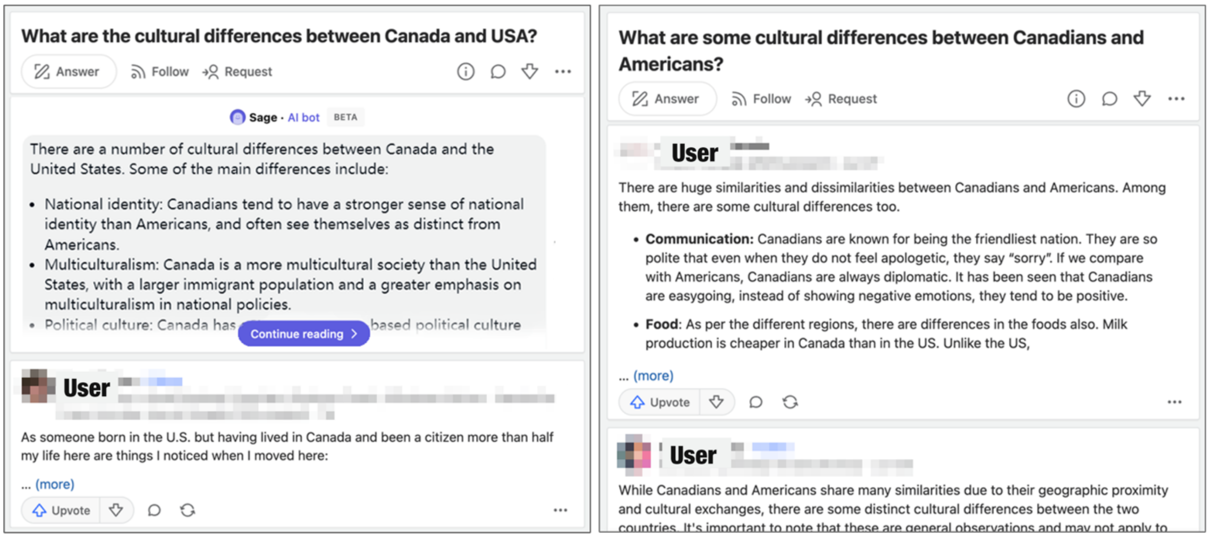
In accordance with the CASA paradigm, we derive concepts of social conformity and differentiation from the field of social science to understand the effects of generative AI answers within our research on knowledge-sharing activities. Specifically, when individuals are presented with generative AI answers, they may either conform to the answers or differentiate themselves from generative AI by contributing answers from a distinctive perspective (Brewer, 1991). We denote the two effects as the "AI-conformity effect" and "AI-differentiation effect." The two effects result in varying degrees of human-AI answer similarity in subsequent human user answers: when the AI-conformity effect dominates, human-AI answer similarity increases; when the AI-differentiation effect dominates, human-AI answer similarity decreases.

3. DATA AND DESCRIPTIVE STATISTICS

Our study uses data from Quora, a leading Q&A platform that boasts more than 400 million monthly active users (Kumar, 2024). The platform covers a wide range of topics, from technology and academic discussions to humanities and personal life experiences, which attract contributors with various interests and areas of expertise (Dey et al., 2021; Kumar, 2024). On January 27, 2023, Quora launched a new policy introducing generative AI as a knowledge provider for some questions on the platform while leaving other questions unaffected (Qureshi, 2023). Specifically, for the questions subjected to the policy, answers provided by generative AI are displayed at the beginning of the answer sections and are distinctly marked as originating from generative AI. For other questions unaffected by this policy, the answer section remains unchanged. Figure 1 presents real-world examples of questions affected by the policy and those unaffected by it. The policy inherently delineates two distinct categories of questions in our analysis: the treatment group, which encompasses questions answered by generative AI after the policy's introduction, and the control group, which comprises questions that have never been addressed by generative AI.

Our observation window for the analysis spans from November 2022 to April 2023, capturing a three-month period both before and after the policy launch (when AI answers are presented on treated question pages) on the platform. To identify questions that receive at least one answer during our pretreatment period, we first extract a random sample of 500,000 activities from this timeframe using the starting and ending identifiers (details on our data collection method and compliance with platform guidelines are available in Appendix B). Then, we keep activities of posting answers and trace back to their original questions. Further, to focus on pre-existing questions, we keep only those questions that were created before the start of our panel data collection (i.e., November 2022). In addition, to ensure the engagement level of the questions in our analysis, we exclude those inactive questions that have fewer than 1,000 views (Claussen et al., 2013). This process leads to the formation of our final main dataset, which comprises 6,878 questions.

The primary focus of our study is to understand the impact of generative AI answers on contributor participation. To measure this participation, we employ two dependent variables: the number of answers and the average answer length in a week. Notably, AI-generated answers are excluded from these measurements. The key independent variable (treatment) in our study is whether a question has received a generative AI answer in a given week. In Table 1, we present definitions and summary statistics of the main variables in our panel data.



(a) a question affected by the policy (b) a question unaffected by the policy

Figure 1: Real-World Examples of Questions Affected and Unaffected by the Policy

Table 1. Definitions of Variables and Summary Statistics

Variable	Definition	Mean	SD	Min	Max
NumAnswer	the number of human answers for a question in a week	0.54	2.67	0	280
AveLenAnswer	the average character count of human answers for a question in a week	98.15	436.08	0	53,782
AIAnswered	whether a question has received the generative AI answer (1: Yes; 0: No) in a week	0.19	0.39	0	1
QuesTenure	the number of months since a question's creation on the platform up to the current week	39.85	33.66	0	158
AccNumAnswer	the accumulated number of answers for a question up to the current week	70.01	365.54	0	20,258
AccNumCollap	the accumulated number of collapsed answers for a question up to the current week	22.42	38.602	0	1,701
LastAnswerInterval	the number of days since the last answer for a question up to the current week	41.47	83.64	0	3,005

Notes: SD is for standard deviation; Min is for minimum; Max is for maximum.

4. EMPIRICAL ANALYSIS

In this section, we conduct our empirical analysis and present the results. We start by exploring the impact of introducing generative AI answers on user participation in knowledge contributions. We then perform various tests to demonstrate the robustness of the main effect. Next, we investigate how generative AI answers affect the content of subsequent human answers, with a focus on the similarity between AI and human answers. To delve deeper, we examine how different types of questions influence the impact of generative AI answers.

Specifically, we find that the question's objectivity and its answer inadequacy moderate the effects of generative AI answers on both contributor participation and content similarity. Finally, our findings indicate that the effects of generative AI answers vary among contributors with different levels of expertise, offering additional insights into the influence of generative AI on human knowledge sharing.

4.1 Main Analysis

In this section, we address our first research question by examining the impact of generative AI answers on contributor participation on the platform. We first employ the two-way fixed effects model to examine the effect of generative AI answers on user knowledge contribution participation. In line with prior research (Liu & Cong, 2023; Shan & Qiu, 2023), we analyze the change in user participation in answering questions after the introduction of generative AI answers using the following difference-in-differences (DID) regression specification:

$$\log(Y_{it}) = \beta_0 + \beta_1 AIAnswered_{it} + X_{it} + \mu_i + \delta_t + \varepsilon_{it} \tag{1}$$

For the dependent variable, Y_{it}, we consider two aspects of contributor participation in question i in week t: the number of answers, denoted as $NumAnswer_{it}$, and the average answer length in terms of character count, denoted as $AveLenAnswer_{it}$. Due to their skewed distribution, we apply a logarithmic transformation for these variables to normalize their scale, enhancing the robustness and interpretability of our regression analysis (Lin et al., 2017; Wang et al., 2022). The independent variable, $AIAnswered_{it}$, serves as a dummy variable indicating whether question i has received the generative AI answer up to week t. The term X_{it} contains a series of control variables: the months since the creation of question i to week t ($QuesTenure_{it}$), the accumulated number of answers to question i by week t ($AccNumAnswer_{it}$), the accumulated number of collapsed answers for question i by week t ($AccNumCollap_{it}$), and days since the last answer to question i by week t ($LastAnswerInterval_{it}$). Finally, we include question fixed effects μ_i to control for potential variations associated with individual question traits. Additionally, we incorporate week fixed effects δ_t to capture time-related fluctuations on the platform over time, such as platform traffic.

The results of the main analysis can be found in Table 2. Columns (1) and (3) respectively display the results of regressions on $NumAnswer_{it}$ and $AveLenAnswer_{it}$ without the inclusion of control variables, while columns (2) and (4) display the results of the regressions that incorporate the controls. Our main interest centers on the coefficients of $AIAnswered_{it}$. Across the results in four columns, the coefficients of $AIAnswered_{it}$ are consistently positive. Specifically, the introduction of the generative AI answer to a question leads to about a 5.5% rise in the number of answers and a 22.5% increase in the average answer length per week.

These findings highlight the significant impact of generative AI answers on user participation in knowledge contribution. In addressing our first research question, our findings add to the understanding of generative AI in knowledge sharing by demonstrating that providing generative AI answers can prompt individuals to participate more on online Q&A platforms.

Table 2. Impact of Generative AI Answers on Contributor Participation

Variables	(1) log(NumAnswer)	(2) log(NumAnswer)	(3) log(AveLenAnswer)	(4) log(AveLenAnswer)
AIAnswered	0.058^{***}	0.055^{***}	0.224^{***}	0.225^{***}
	(0.007)	(0.007)	(0.028)	(0.028)
Control variables	No	Yes	No	Yes
Question fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Observations	171,950	171,950	171,950	171,950

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

4.2 Robustness Checks

In the preceding section, we demonstrate that introducing generative AI answers encourages subsequent human knowledge contributions. To reinforce the robustness of our conclusions, we undertake a series of checks by employing various empirical methods. Table 3 provides a summary of these checks.

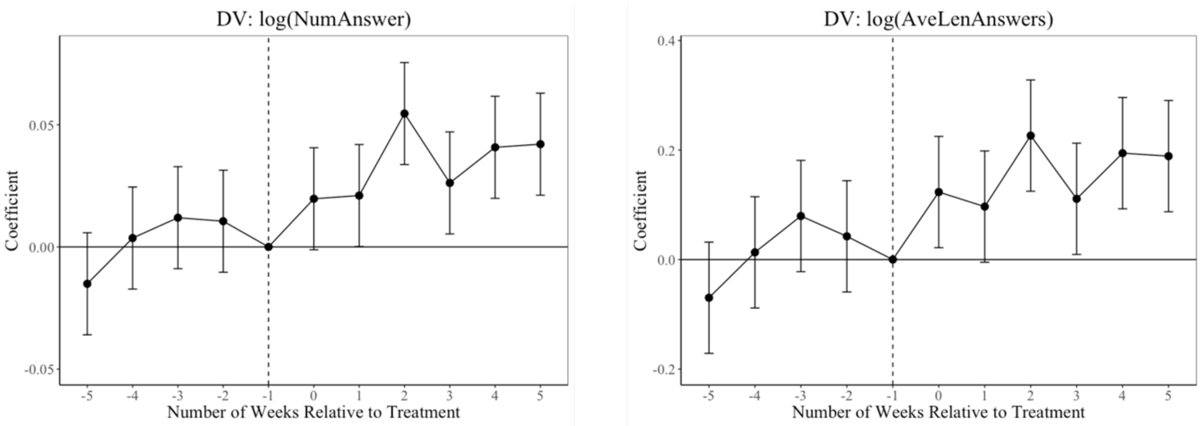
Table 3. Overview of Robustness Checks

Empirical Methods	Description	Primary Takeaways
Relative Time Model (RTM)	We employ RTM to ensure that the assumption of parallel trends holds.	The assumption of parallel trends holds, meaning that pretreatment trends do not confound the observed effect.
Coarsened Exact Matching (CEM)	We employ CEM to address the potential impact of heterogeneity between the treatment and control groups.	Our results remain robust after adjusting for the selection bias related to observable factors.
Analysis with the Content-Matched Question Subsample	We create a subsample of questions that are matched by similar question content so that the control and treatment units are more comparable.	The results are consistent with those observed in the main analysis, which further indicates that our main findings are not affected by the potential issue of selection bias.
Excluding the Potential Impact of Existing Answers	We analyze a dataset containing only questions created within one week before the policy introduction to exclude the potential impact of existing answers.	Our results remain robust after excluding the potential impact of existing answers.
Excluding the Potential Impact of External Generative AI Tool Use	We analyze a dataset excluding answers that are detected as AI-generated to exclude the potential impact of contributions using external generative AI tools.	Our results remain robust after excluding the potential impact of external generative AI tool use.

4.2.1 Relative Time Model

Following prior research (Autor, 2003), we utilize the RTM approach to determine whether trends in knowledge contribution participation are parallel between the treatment and control groups prior to the introduction of the generative AI answers. If the relative time terms prior to AI answer inclusion are insignificant, we can reckon that the treatment and control groups are comparable in the absence of treatment and attribute the differences in outcomes between the two groups after AI answer inclusion to the average treatment effect (Angrist & Pischke, 2008). Specifically, we establish Equation (2) by replacing $AIAnswered_{it}$ in Equation (1) with a series of dummy variables, $Week_{it}^{\tau}$, to include indicators for five weeks preceding and following the treatment. In Equation (2), $\tau \in \{-5, \ldots, -2, 0, \ldots, 5\}$. The last pretreatment week ($\tau = -1$) serves as the baseline group. If week t represents the τth week relative to the baseline group, $Week_{it}^{\tau}$ = 1; otherwise, $Week_{it}^{\tau}$ = 0.

$$\log(y_{it}) = \beta_0 + \sum \beta_\tau Week_{it}^{\tau} + X_{it} + \mu_i + \delta_t + \varepsilon_{it} \tag{2}$$



Note: Error bars show 95% confidence intervals.

Figure 2. Impact of Generative AI Answers on Contributor Participation Over Time

The parameters β_{-5} to β_{-2} are the coefficients of lead indicators, representing the week-to-week shifts in pretreatment periods. Therefore, if the parallel trends assumption is satisfied, the coefficients of leading indicators should not be significant. Figure 2 visually displays the results of the RTM analysis. Notably, the coefficients of lead indicators are all insignificant, suggesting the parallel trend assumption holds in our main analysis. Consequently, this RTM analysis effectively addresses concerns about non-parallel pretreatment trends between treatment and control groups as a confounder (Li et al., 2022).

4.2.2 Coarsened Exact Matching

In this section, we adopt the CEM method (Guo et al., 2023; Nian et al., 2021) to mitigate potential selection bias concerns. It is worth noting that even though we use the DID method, which can address some selection bias with the employment of fixed-effects settings, our conclusions are not exempt from endogeneity concerns. In our specific context, the decision to display a generative AI answer, which determines whether a question belongs to the treatment or control group in our analysis, is made by the platform. Consequently, a primary concern regarding endogeneity is the non-random selection of questions to display generative AI answers. One could argue that inherent differences may exist between our treatment and control groups. In comparison with propensity score matching (PSM), the CEM method enables researchers to independently and precisely match groups across various attributes (Dewan et al., 2023; Mayya et al., 2021). The observations will be categorized into strata based on a set of observable covariates. The matched dataset is then generated by containing only strata containing both treated and control observations (Aggarwal & Hsu, 2014).

For matching, we include a series of variables, wherein time-varying covariates are measured using their pretreatment values from the period immediately before the policy introduction: the months since the creation of the question before the policy ($QuesTenure$), the accumulated number of answers to the question before the policy ($AccNumAnswer$), the accumulated number of collapsed answers for the question before the policy ($AccNumCollap$), days since the last answer to the question before the policy introduction ($LastAnswerInterval$), the number of question followers ($QuesFollow$), the number of views ($QuesView$), whether the question was raised by a user awarded a Top Writer badge by Quora ($QuesFromTop$; Bodnick, 2012), and the Standard Simple Measure of Gobbledygook (SMOG) index ($QuesSmog$) to assess the question's readability (Ghose et al., 2012; Xu et al., 2021). Additionally, we include a dummy indicator of question openness ($QuesOpen$), classified using OpenAI's GPT-4 model, with detailed prompt instructions provided in Appendix C.

We employ the "MatchIt" package in R (Stuart et al., 2011) with the "k2k" option to perform an exact one-to-one matching within the strata. To examine the matching quality, we also carry out t-tests to compare variable means before and after matching. The variable-to-variable comparison is presented in Table 4. Notably, after matching, the differences for all covariates between the treatment and control groups become insignificant, suggesting that the treatment and control groups are appropriately matched and that the covariate imbalance has been greatly reduced.

Table 4. Summary Statistics Before and After CEM

Variables	Before Matching			After Matching		
	Mean (Treated)	Mean (Control)	Difference	Mean (Treated)	Mean (Control)	Difference
QuesTenure	49.518	34.194	15.324***	45.178	44.250	0.928
AccNumAnswer	66.618	72.533	-5.915	44.957	47.989	-3.032
AccNumCollap	22.492	22.237	0.255	20.014	19.031	0.983
LastAnswerInterval	44.663	41.589	3.074	38.669	37.687	0.982
QuesFollow	20.450	17.789	2.661	12.281	12.243	0.038
QuesLen	66.695	78.288	-11.593***	63.870	63.776	0.094
QuesView	266435.8	190120.6	76315.2**	75910.02	68275.19	7634.83
QuesOpen	0.612	0.655	-0.043***	0.595	0.590	0.005
QuesFromTop	0.016	0.013	0.003	0.002	0.002	0.000
QuesSmog	7.891	8.279	-0.388***	7.676	7.685	-0.009

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We then re-examine DID regressions from Equation (1) using the matched sample and present the results in Table 5. These results are consistent with our primary results derived from the full-sample dataset. Therefore, by creating a more comparable control group, this analysis further supports the positive effect of introducing generative AI answers on contributor participation.

Table 5. Impact of Generative AI Answers after CEM

Variables	(1) log(NumAnswer)	(2) log(AveLenAnswer)
AIAnswered	0.027***	0.092***
	(0.009)	(0.035)
Control variables	Yes	Yes
Question fixed effects	Yes	Yes
Week fixed effects	Yes	Yes
Observations	92,300	92,300

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

4.2.3 Analysis with the Content-Matched Question Subsample

In this section, we create a subsample of questions that are matched by essentially similar question content. For example, consider the questions "How can I loosen my hip flexors?" and "What are the best stretches for hip flexors?". While they are phrased in slightly different ways, both of them essentially seek methods to relax hip flexors. The previous question belongs to the control group that receives no generative AI answer. Conversely, the latter question belongs to the treatment group that receives answers from the generative AI. For this specific question, namely, "What are the best stretches for hip flexors?", the generative AI answer shown in the answer section offers a detailed illustration of three examples of stretches for hip flexors. Due to their content similarity, these two questions will be grouped as a matched question pair. To systematically identify such pairs, we first employ the Sentence-BERT (SBERT) method, which is designed to calculate embeddings for sentences or short paragraphs in a way that the cosine similarity between the embeddings of two sentences or paragraphs reflects their semantic similarity (Reimers & Gurevych, 2019). Cosine similarity scores range from -1 to 1, with higher values indicating greater semantic similarity. We compute the SBERT-based semantic similarity matrix for all questions in our dataset based on their textual content and identify question pairs with a similarity score greater than 0.7. We then manually review these pairs and retain those where the questions are semantically equivalent, ensuring that they are essentially asking the same thing. Next, we group the retained pairs into matched sets, with each resulting set containing at least one treated question and 1 to 4 semantically equivalent control questions. This process results in 191 distinct question sets, which constitute the subsample used for our analysis. Using this content-matched question subsample, we re-estimate the regression model in Equation (1). The results are presented in Table 6. Notably, the results are consistent with the results observed in the main analysis using the full-sample dataset, further suggesting that our main findings are not affected by the potential issue of selection bias.

Table 6. Main Effect Analysis with Content-Matched Question Subsample

Variables	(1) log(NumAnswer)	(2) log(AveLenAnswer)
AIAnswered	0.054^{***}	0.211^{***}
	(0.014)	(0.065)
Control variables	Yes	Yes
Question fixed effects	Yes	Yes
Week fixed effects	Yes	Yes
Observations	18,825	18,825

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

Additionally, with this content-matched question dataset, we perform a *t*-test evaluating the balance between the treatment and control groups across the variables used in our matching process. The results suggest no statistical difference between the two groups, with details available in Appendix D. This finding suggests a high degree of randomness in the selection of questions for treatment or control groups, which further alleviates concerns about the potential selection bias.

4.2.4 Excluding the Potential Impact of Existing Answers

Some might contend that pre-existing answers could potentially exert influence on the effects of the generative AI answers, leading to endogeneity concerns caused by disparities in pretreatment answers between the control and treatment groups. To further address these concerns, we conduct an analysis using an additionally collected dataset. Specifically, we collect a random 25,000 questions posed within one week before the policy's inception (i.e., from January 20 to January 27, 2023), and keep those that received at least one answer during the three-month post-policy observation window. The recentness of the questions in the dataset ensures minimal interference from previously existing answers. Since there is no pretreatment observation in this analysis, we do not include question fixed effects in the analysis, leading to the following regression specification:

$$\log(Y_{it}) = \beta_0 + \beta_1 AIAnswered_{it} + X_{it} + \delta_t + \varepsilon_{it} \tag{3}$$

The results of our analysis show that the coefficients for $AIAnswered_{it}$ are significantly positive for both the number of answers and the average length of answers, which are consistent with our main results. With the introduction of generative AI answers, we observe an increase in the number of answers and the average answer length. Table 7 displays detailed results. These results demonstrate the robustness of our conclusions when the potential impact of existing answers is excluded.

Table 7. The Effect Excluding the Potential Impact of Pretreatment Answers

Variables	(1) log(NumAnswer)	(2) log(AveLenAnswer)
AIAnswered	0.046***	0.079***
	(0.010)	(0.012)
Control variables	Yes	Yes
Week fixed effects	Yes	Yes
Observations	163,995	163,995

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

4.2.5 Excluding the Potential Impact of External Generative AI Tool Use

One potential concern regarding the validity of our findings is the possibility that contributors may use external generative AI tools (e.g., ChatGPT) to produce answers, especially when AI-generated answers are displayed on the question pages. To mitigate the concern, we exclude answers that could potentially be generated by AI and re-estimate our analysis. To determine whether answers attributed to human contributors are influenced by external AI tools, we adopt two AI content detection tools. First, we utilize a GPT output detection tool developed by OpenAI, as described in prior research (Cingillioglu, 2023; Perkins et al., 2024; Shan & Qiu, 2023). After excluding answers detected as AI-generated, we apply Equation (1) to the dataset, and the analysis results are robust and are presented in Columns (1) and (2) of Table 8. Second, we employ GPTZero, developed by Edward Tian, which is specifically designed to detect plagiarism and AI-generated content from the AI bot ChatGPT (Elkhatat et al., 2023). For each sentence, we use GPTZero to assign a probability that it is generated by an AI. We exclude an answer if the average probability across its sentences exceeds 0.5, and then re-apply Equation (1) to this dataset. The estimation results from this analysis are displayed in Columns (3) and (4) of Table 8. These results are consistent with our main results, which collectively suggest that our main findings are robust when the potential impact of external generative AI tool use is removed.

Table 8. The Effect Excluding the Potential Impact of External Generative AI Tool Use

Variables	(1)	(2)	(3)	(4)

	log(NumAnswer)	log(AveLenAnswer)	log(NumAnswer)	log(AveLenAnswer)
AIAnswered	0.040***	0.210***	0.016***	0.126***
	(0.006)	(0.025)	(0.002)	(0.016)
Control variables	Yes	Yes	Yes	Yes
Question fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Observations	171,950	171,950	171,950	171,950

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

4.3 The Effect on the Content of Human Contributions

In this section, we investigate our second research question: *How does the introduction of generative AI answers influence the content of subsequent human answers?* Specifically, we focus on the content similarity between human and AI answers. To explore this, we utilize the content-matched question subsample as described in Section 4.2.3, where questions within a set are all asking the same thing.

For each question within a matched question set, we evaluate the similarity between human and AI answers. Specifically, if a question is from the treatment group (a generative AI answer is provided), the answer similarity is measured against the AI answer for that question. Conversely, if a question is from the control group, its human answer is compared against the AI answer for a counterpart question from the treatment group within the same set. For sets containing multiple questions from the treatment group (i.e., multiple AI answers are available for comparison), the human answers for control group questions within the set are compared against a randomly selected AI answer from the same set.

To enhance the robustness of our results, we utilize two different methods to measure the human-AI answer similarity. First, we leverage OpenAI's GPT-4 model to rate the similarity between a human answer and an AI answer from 0% to 100%, with 0% being "totally different" and 100% being "exactly the same" (OpenAI, 2023a; for detailed prompt instructions, see Appendix C). Second, we apply the SBERT method to compute semantic similarity scores based on the cosine similarity between sentence embeddings of human answers and AI answers.

The following regression specification enables us to analyze the effects of generative AI answers on answer similarity:

$$Z_{ijt} = \beta_0 + \beta_1 AIAnswered_{ijt} + X_{ijt} + \xi_j + \delta_t + \varepsilon_{ijt} \quad (4)$$

In the regression equation, Z_{ijt} denotes the average similarity of human answers for question i in set j during time period t against the generative AI answer (referred to as "human-AI answer similarity"). It includes two measures of human-AI answer similarity, $AveSimGPT_{ijt}$ and $AveSimSBERT_{ijt}$, respectively derived with GPT-4 and SBERT methods. The vector of control variables is denoted as X_{ijt} in Equation (7). Apart from week fixed effects, δ_t, we also include set fixed effects, ξ_j, to control for potential variations among matched sets.

The analysis results are presented in Table 9. Columns (1) and (2) respectively display the results of regressions on $AveSimGPT_{ijt}$ $and\, AveSimSBERT_{ijt}$. Regardless of the measurement approach—be it GPT-4 or SBERT—the coefficients of $AIAnswered_{ijt}$ are consistently positive, supporting that the introduction of generative AI answers leads to a rise in human-AI answer similarity.

These results answer our second research question by suggesting that the introduction of generative AI answers motivates users to provide answers that align more closely with AI-generated content, highlighting the AI-conformity effect as the underlying mechanism for the main effect. This finding reflects informational conformity, where individuals adjust their behavior or opinions based on information from others that they perceive as reliable knowledge (Deutsch & Gerard, 1955). In this context, contributors align their answers with generative AI as a credible source of information.

Table 9. Impact of Generative AI Answers on Human-AI Answer Similarity

Variables	(1) AveSimGPT	(2) AveSimSBERT

AIAnswered	0.074***	0.041***
	(0.024)	(0.013)
Control variables	Yes	Yes
Set fixed effects	Yes	Yes
Week fixed effects	Yes	Yes
Observations	2,049	2,049

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; Robust standard errors clustered at the set level are in parentheses.

4.4 Moderation Analyses

In this section, we investigate our third research question: *Does the introduction of generative AI answers have heterogeneous effects on different types of questions*? Specifically, we explore how the effects of generative AI answers on contributor participation are influenced by the question's objectivity and answer inadequacy.

4.4.1 Question's Objectivity

First, we investigate whether the question's objectivity influences the effect. As posited earlier, we aim to examine whether objective questions (vs. subjective questions) are more impacted by the introduction of generative AI answers. We utilize OpenAI's GPT-4 model to differentiate questions into objective or subjective categories, labeling the objective ones as 1 and the subjective ones as 0. We additionally require GPT-4 to provide its confidence level of the judgment on a scale of 1 to 9, with 1 being "not at all confident" and 9 being "extremely confident" (for detailed prompt instructions, see Appendix C). To ensure the consistency of GPT-4's responses to our inquiries, we access the model on three separate dates using three unique accounts. The categorization of questions and confidence levels remain consistent across the requests. We incorporate the question's objectivity as a moderating factor and formulate the regression models accordingly:

$$\log(Y_{it}) = \beta_0 + \beta_1 AIAnswered_{it} + \beta_2 AIAnswered_{it} \times QuesObjective_i + X_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (5)$$

where $QuesObjective_i$ denotes whether question i is objective or subjective (1 = objective, 0 = subjective). The results are presented in Columns (1) and (2) of Table 10. The positive coefficients of the interaction between $AIAnswered$ and $QuesObjective$ suggest a greater positive influence of the generative AI answers on objective questions relative to subjective ones. The results imply that when a generative AI answer is displayed, objective questions receive a greater positive impact than subjective questions in terms of the number of answers and the average answer length. We further conduct the analysis in Equation (5) using a subset of questions that the GPT-4 model is "extremely confident" in categorizing as either objective or subjective. The results are consistent (for detailed results, see Appendix E).

The analysis regarding the moderating role of the question's objectivity offers converging evidence for our proposed mechanism. As previously theorized, given that people tend to perceive AI as more trustworthy in objective tasks in comparison to subjective tasks (Castelo et al., 2019), they are more inclined to conform to generative AI answers as a credible information source when the question is objective than when it is subjective (Cialdini & Goldstein, 2004; Deutsch & Gerard, 1955). If the positive effects of generative AI answers on subsequent contributor participation are primarily driven by the AI-conformity effect (vs. AI-differentiation effect), then the positive effects of generative AI answers should be more pronounced among objective questions compared with subjective questions. Therefore, the positive coefficients of the interaction between $AIAnswered$ and $QuesObjective$, indicate greater effects of generative AI answers for objective (vs. subjective) questions and provide empirical support for the AI-conformity effect as the underlying mechanism.

Table 10. Moderating Effects of Question Objectivity and Answer Inadequacy

Variables	(1) log(NumAnswer)	(2) log(AveLenAnswer)	(3) log(NumAnswer)	(4) log(AveLenAnswer)
AIAnswered	0.043^{***} (0.009)	0.154^{***} (0.039)	0.032^{***} (0.012)	0.159^{***} (0.045)
AIAnswered × QuesObjective	0.030^{***} (0.011)	0.143^{***} (0.045)		
AIAnswered × UnderAnswered			0.049^{**} (0.011)	0.122^{***} (0.047)
Control variables	Yes	Yes	Yes	Yes
Question fixed effects	Yes	Yes	Yes	Yes

Week fixed effects	Yes	Yes	Yes	Yes
Observations	171,950	171,950	171,950	171,950

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

To further examine our theorizing, we employ the content-matched question subsample generated previously for mechanism analysis. With this dataset, we carry out additional analysis with the question's objectivity as the moderator and human-AI answer similarity as the dependent variable to generate the following specifications:

$$Z_{ijt} = \beta_0 + \beta_1 AIAnswered_{ijt} + \beta_2 AIAnswered_{ijt} \times QuesObjective_i + X_{ijt} + \xi_j + \delta_t + \varepsilon_{ijt} \quad (6)$$

As in Equation (4), Z_{ijt} denotes the average human-AI answer similarity for question i in set j during time period t. The results are presented in Columns (1) and (2) of Table 11. The positive coefficients of the interaction between $AIAnswered$ and $QuesObjective$ indicate that the increase in human–AI answer similarity following the introduction of generative AI answers is greater for objective questions than subjective questions. The findings further underscore the AI-conformity as the underlying mechanism, showing that human users conform more to generative AI answers in domains where AI is perceived to provide reliable information (i.e., objective questions) than in domains where AI is not traditionally seen as highly capable (i.e., subjective questions).

Table 11. The Effect on Human-AI Answer Similarity: Moderating Effects of Question Objectivity and Answer Inadequacy

Variables	(1) AveSimGPT	(2) AveSimSBERT	(3) AveSimGPT	(4) AveSimSBERT
AIAnswered	0.052^{**} (0.023)	0.041^{***} (0.015)	0.063^{**} (0.026)	0.039^{***} (0.014)
AIAnswered × QuesObjective	0.080^{**} (0.034)	0.039^{**} (0.019)		
AIAnswered × UnderAnswered			0.091^{**} (0.046)	0.061^{**} (0.027)
Control variables	Yes	Yes	Yes	Yes
Set fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Observations	2,049	2,049	2,049	2,049

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$; Robust standard errors clustered at the set level are in parentheses.

4.4.2 Question's Answer Inadequacy

We then delve into whether the impact of generative AI answers varies with different levels of answer inadequacy for questions. Specifically, we introduce a dummy variable, $UnderAnswered$, which categorizes questions based on the number of their answers prior to the policy launch. A value of "1" represents questions in the bottom 50% concerning the answer volume obtained prior to the policy, signifying that they are "under-answered." Conversely, a value of "0" represents questions in the top 50% concerning the answer volume obtained prior to the policy, signifying that they are "well-answered." We replace $QuesObjective_i$ with $UnderAnswered_i$ as the moderating factor in Equation (5), $UnderAnswered_i$ denotes whether question i is under-answered or well-answered (1 = under-answered, 0 = well-answered). Columns (3) and (4) in Table 10 display the results of this analysis. The positive coefficients of the interaction between $AIAnswered$ and $UnderAnswered$ suggest that the influence of the generative AI answers on contributor participation (including the number and average length of answers) is greater for under-answered questions than well-answered questions.

This finding aligns with our explanation of the main effect: informational conformity tends to be stronger when individuals face uncertainty or lack sufficient guidance from others (Cialdini & Goldstein, 2004; Claidière & Whiten, 2012). Under-answered questions, by definition, offer limited peer input, making it more challenging for contributors to figure out what constitutes a credible or contextually appropriate answer from the existing content. In such contexts, a generative AI answer can serve as a salient source of credible information. This reduces uncertainty and encourages participation through informational conformity (Deutsch & Gerard, 1955; Cialdini & Goldstein, 2004). In contrast, well-answered questions already contain ample informational input from peers, diminishing the relative informational value of an AI-generated answer (Lindberg et al., 2022). As such, the more pronounced effect among under-answered questions provides converging empirical support for the AI-conformity mechanism as the explanation for how generative AI answers influence contributor participation.

Moreover, we replace the moderator in Equation (6) with the answer inadequacy dummy and re-estimate the effects on human-AI answer similarity. The results are presented in Columns (3) and (4) of Table 11. The coefficients for the interaction between $AIAnswered$ and $UnderAnswered$ are consistently positive, regardless of whether the GPT-4 model or the SBERT methodology is used. This indicates that the introduction of generative AI answers increases the similarity between human and generative AI answers more for under-answered questions than for well-answered ones. These findings provide additional support for the AI-conformity effect by suggesting that contributors are more likely to conform their answers to the AI-generated answer when peer input is limited.

4.5 The Effect on Human Experts' and Non-experts' Contribution

In this section, we further investigate our fourth research question: *How does the introduction of generative AI answers affect the participation of contributors with varying levels of expertise?* In our research context, we incorporate data from Quora's "Top Writer" badge, which is awarded to contributors based on the quality of their responses (Bodnick, 2012). This allows us to differentiate between expert and non-expert contributors for a more nuanced analysis. Specifically, contributors with the Top Writer badge on Quora are identified as human experts while those without the badge are identified as non-experts. We use $NumExpert$ and $NumNonExpert$ to represent the number of answers from experts and the number of answers from non-experts, respectively, and perform the following regressions:

$$\log(NumExpert_{it}) = \beta_0 + \beta_1 AIAnswered_{it} + X_{it} + \mu_i + \delta_t + \varepsilon_{it} \tag{7}$$

$$\log(NumNonExpert_{it}) = \beta_0 + \beta_1 AIAnswered_{it} + X_{it} + \mu_i + \delta_t + \varepsilon_{it} \tag{8}$$

The results are presented in Table 12. Our findings reveal that the coefficients for $AIAnswered_{it}$ are positive for both expert and non-expert contributors, indicating that AI-generated answers encourage participation from both groups. However, the impact of generative AI answers is more pronounced among non-expert contributors. Specifically, the introduction of AI-generated answers leads to approximately a 0.7% increase in responses from experts and a 5.6% increase in responses from non-experts.

Prior research on informational conformity has shown that individuals with lower subjective knowledge are more likely to conform to external information sources (Lascu et al., 1995). In our context, non-experts, who often lack deep domain knowledge, are more likely to perceive the AI-generated answer as a helpful informational source when deciding whether and how to contribute (Rios et al., 2018). In contrast, experts are better equipped with substantial relevant knowledge, which reduces the influence of AI-generated answers (Logg et al., 2019). As such, the stronger effect among non-experts reinforces our explanation that the AI-conformity effect underlies the influence of generative AI answers on contributor participation. These results demonstrate a clear distinction in how AI-generated content influences user behavior based on expertise level. The stronger effect among non-expert users suggests that AI-generated answers can play an important role in lowering barriers to participation, particularly for individuals who may otherwise lack the confidence or knowledge to contribute. This differential impact underscores the role of AI as an enabler of broader knowledge sharing.

Table 12. Impact of Generative AI Answers on the Number of Human Expert and Non-Expert Contributors

Variables	(1) log(NumExpert)	(2) log(NumNonExpert)
AIAnswered	0.007^{***}	0.056^{***}
	(0.001)	(0.007)
Control variables	Yes	Yes
Question fixed effects	Yes	Yes
Week fixed effects	Yes	Yes
Observations	171,950	171,950

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

5. RANDOMIZED EXPERIMENT

We further conduct an additional randomized experiment to complement the empirical analysis of data from the leading online Q&A platform. The experiment has two objectives. Primarily, we aim to provide stronger causal evidence to complement the empirical findings. One concern regarding the causal identification of our natural experiment pertains to the potential violation of SUTVA. Specifically, while our empirical analysis operates at the question level, contributors on the Q&A platform may encounter both questions with and without AI-generated answers. In our controlled experiment, we mitigate this concern by randomizing at the participant level and leveraging the distribution features of the Prolific Academic platform to limit participant exposure to a single condition. Through this well-controlled design in our randomized experiment, we also further address potential selection biases in a more direct manner by employing an identical set of questions across various conditions and randomly assigning participants to each condition. Secondly, we aim to provide further evidence for the AI-conformity effect by directly measuring participants' conformity behavior when formulating their own answers, thereby offering deeper insight into the mechanism underlying the observed effects.

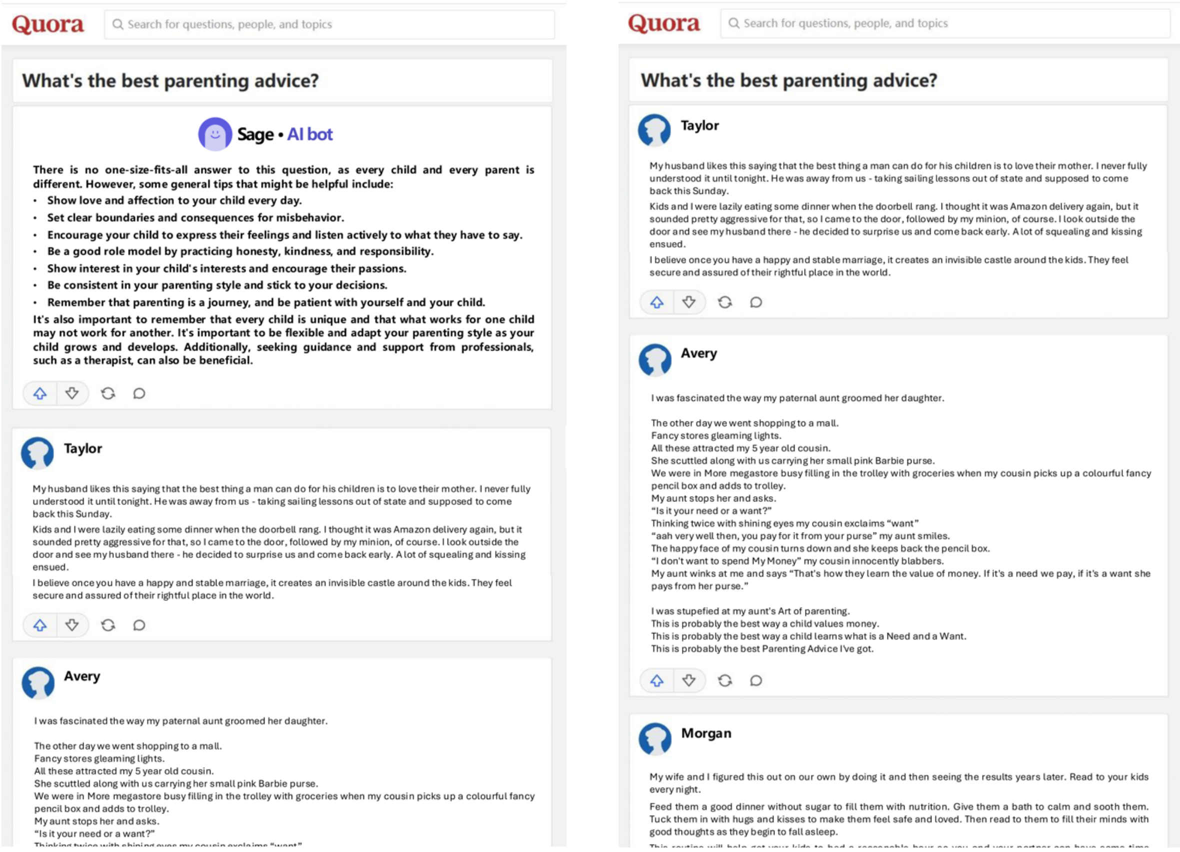
5.1 Method

The experiment employs a 2-cell (treatment vs. control) between-subjects design. We manipulate the independent variable by presenting participants with question pages with or without generative AI answers. In the treatment condition, participants view a question page where the generative AI answer is positioned at the top of the answer section. The generative AI answer is marked with an avatar of "Sage" and the label "Sage · AI bot." Below the generative AI answer, three answers from users are displayed, each accompanied by a cartoon avatar and a fake human-like name. In the control condition, participants view the same question page without a generative AI answer. The question page displays the same three answers from users, presented in the same order and with identical avatars and names as in the treatment condition. To better mimic real online Q&A platforms, we also include elements such as a search bar, upvote and downvote buttons, a forward button, and a comment button, which are common on Q&A platforms. The two conditions differ only in the presence or absence of the generative AI answer at the top of the answer section; all other aspects of the page layout are held constant. To improve the generalizability of our experimental results, we include 10 distinct questions to reduce the potential influence of any single question's characteristics. All questions and answers used in the randomized experiment are extracted directly from our empirical dataset. Table 13 presents sample question pages that we utilize in the randomized experiment.

A total of 200 participants (45.0% female, M_{age} = 37.08 years, SD = 12.56) who have experience in using Q&A platforms are recruited from Prolific Academic and randomly assigned to one of the 20 experimental groups, resulting from crossing 2 conditions (treatment vs. control) with 10 different questions. All participants are paid with proper monetary compensation upon completion. After confirming their qualification of being a Q&A platform user and providing their consent to participate, the participants are directed to a question page. They are instructed to imagine themselves browsing Quora and coming across the question. To better align their incentive to contribute with real-world scenarios, participants are also informed that the questions and existing answers are directly sourced from Quora. In addition, they are explicitly informed that they are not permitted to consult large language models such as ChatGPT during the experiment.

Table 13. A Sample of Question Pages in the Experiment

Treatment condition	Control condition



Participants first view the question page, which displays a question followed by existing answers. Afterward, participants are directed to an answering page where they are instructed to write their own answers. On the answering page, they can click a "*Previous Page*" button, which allows them to return to the earlier page displaying the question and existing answers. Participants are not required to revisit the previous page, but they are free to do so. They may navigate between the answering page and the previous page multiple times.

We record the cumulative amount of time (in seconds) that each participant spends on the previous page after first returning to it. This total duration serves as a behavioral proxy for *conformity*. This proxy is based on the premise that revisiting and spending time viewing existing answers reflects a behavioral tendency to seek informational input from the surrounding environment when formulating one's own answer. Extended time spent on the previous page suggests a greater tendency to seek and attend to external information, consistent with behaviors associated with informational conformity (Deutsch & Gerard, 1955; Cialdini &

Goldstein, 2004).

By tracking participants' viewing behavior during the answer generation process, this measure provides an unobtrusive and behaviorally grounded indicator of conformity and offers a suggestive empirical test of our central mechanism, the AI-conformity effect: In our experimental design, the sole difference between the treatment and control conditions is the presence of an AI-generated answer on the preceding page. We therefore record and compare the total time participants in each condition spend re-reviewing that page. Should the treatment group devote significantly more seconds to revisiting existing responses than the control group, we can infer that this extra dwell time reflects engagement with the AI answer itself. In other words, because every other aspect of the task environment is held constant, the additional time investment by treated participants constitutes a clear behavioral signature of their effort to read, process, and integrate AI-generated content, which provides a direct measure for our key mechanism, AI-conformity effect.

Participants subsequently indicate their familiarity with the topic of the presented question (1 = not at all, 9 = very) and their frequency of answering questions on Q&A platforms, selecting from seven options ranging from "Less than once a month" to "Several times a day," with values spanning from 1 to 7. To verify that participants pay attention to the content, we include an attention check asking participants to identify the question they have been presented with during the experiment.

5.2 Results

5.2.1 Answer Length

Consistent with our analysis using Quora data, we measure human contributor participation in knowledge sharing by the length of answers provided by participants. We first conduct a linear mixed-effects analysis with answer length as the dependent variable, treatment as the independent variable, familiarity with the question topic and frequency of answering questions on Q&A platforms as covariates, and question ID included as a random factor. The results reveal a significant main effect of treatment ($\beta = 0.271$, $SE = 0.091$, $t(185.8) = 2.975$, $p = 0.003$). Specifically, participants in the treatment condition (M = 5.593, SE = 0.066) provide significantly longer answers than participants in the control condition (M = 5.330, SE = 0.068). These findings indicate that the introduction of generative AI answers significantly increases the length of participants' answers to a question, consistent with results obtained from our analysis of Quora data.

To examine whether conformity behavior (measured by the total time participants in each condition spend re-reviewing the page of displaying the question and existing answers, as discussed earlier) mediates the effect of the generative AI answer on answer length (our key mechanism), we conduct a mediation analysis with 5,000 bootstrap samples. The results reveal that treatment significantly predicts conformity behavior ($\beta = 0.695$, $SE = 0.304$, $p = 0.023$), suggesting that the treatment group spends significantly more time revisiting existing responses than the control group. Since the inclusion of an AI-generated answer is the only difference between conditions, this additional dwell time can be interpreted as a direct measure of participants' engagement with that AI content, thereby making the AI-conformity effect a more plausible explanation.

In addition, conformity behavior significantly predicts answer length ($\beta = 0.081$, $SE = 0.021$, $p < 0.001$). The indirect effect of treatment on answer length through conformity behavior is significant ($\beta = 0.056$, 95% CI [0.008, 0.120]). These findings suggest that the introduction of generative AI answers increases participants' conformity behavior, which in turn enhances their participation, further supporting the AI-conformity effect as the key mechanism.

5.2.2 Human-AI Answer Similarity

We analyze human-AI answer similarity in this randomized experiment to further examine the

AI-conformity effect. Consistent with Section 4.3, we measure similarity using two methods: OpenAI's GPT-4 model and the SBERT model. We then perform a linear mixed-effects analysis with human-AI answer similarity as the dependent variable, keeping the same set of independent variables, covariates, and random factor as in the previous analysis of answer length. Separate analyses are conducted for each similarity measurement method.

The results reveal a significant main effect of treatment across both measurement methods (β_{GPT} = 0.148, SE_{GPT} = 0.032, t_{GPT} (186.5) = 4.574, p_{GPT} < 0.001; β_{SBERT} = 0.050, SE_{SBERT} = 0.021, t_{SBERT} (185.7) = 2.419, p_{SBERT} = 0.017). Specifically, participants in the treatment condition provide answers that are more similar to generative AI answers (M_{GPT} = 0.585, SE_{GPT} = 0.024; M_{SBERT} = 0.663, SE_{SBERT} = 0.016) compared to participants in the control condition (M_{GPT} = 0.436, SE_{GPT} = 0.023; M_{SBERT} = 0.604, SE_{SBERT} = 0.016). These findings indicate that the presence of generative AI answers motivates human users to contribute answers that align more closely with those generated by AI, supporting the AI-conformity effect.

5.3 Discussion

Based on the well-controlled experimental paradigm, our results offer direct causal support for the positive impact of introducing generative AI answers on human contributor participation in knowledge sharing. We observe significant increases in both participants' participation in knowledge sharing, reinforcing our empirical findings from Quora.

Importantly, this experimental design allows us to directly test the AI-conformity mechanism underlying the observed effects of generative AI answers. Specifically, we assess participants' conformity behavior when they construct their own answers. The analyses support the mediating role of conformity in the effects of generative AI answers on participants' answer length. Moreover, we analyze human-AI answer similarity in the experimental data and find that participants tend to align their answers to the generative AI answer when it is present, consistent with findings from Quora data. These converging results provide supporting evidence for the role of AI conformity in explaining why the introduction of generative AI answers increases human contributor participation.

6. GENERAL DISCUSSION AND CONCLUSIONS

Recent advancements in generative AI enable it to provide responses in human-like languages, suggesting the potential of generative AI as a knowledge provider in knowledge-sharing activities (Dwivedi et al., 2023). Yet, concerns about drawbacks pertaining to incorporating generative AI into knowledge-sharing communities still persist (Susarla et al., 2023). This research provides valuable insights into the understanding of the effects of generative AI on knowledge-sharing dynamics.

Leveraging Quora's unique policy that introduces generative AI as a knowledge provider on the platform, our analysis reveals that generative AI answers positively influence subsequent human contributor participation, increasing both answer quantity and length. To further explore these effects, we examine the content of subsequent human answers and observe an increase in human-AI answer similarity, which provides converging evidence for the AI-conformity effect as the underlying mechanism. Our findings also indicate that the positive influence of generative AI is more pronounced for objective and under-answered questions, and is stronger among non-expert contributors compared to expert ones.

6.1 Theoretical Contribution and Managerial Implications

To our knowledge, this research presents the first empirical investigation into the impact of incorporating generative AI as a direct knowledge provider. Our study complements previous work by focusing on the context of embedded AI tools, which operate differently from external tools like ChatGPT (i.e., Burtch et al., 2024; Li & Kim, 2024; Sanatizadeh et al., 2023; Xue et al., 2023). Our findings indicate that embedded generative AI can potentially increase user participation by serving as a credible information source in a seamless manner. This integration allows the AI to operate within the context of specific tasks, workflows, or user interactions, enhancing engagement and user contributions. First, embedded generative AI can leverage specific context and data from the platform it resides in. This allows it to generate responses that are more tailored and relevant to the users' immediate needs, enhancing the quality of knowledge sharing (Olan et al., 2022; Puntoni et al., 2021). Second, by being integrated into existing platforms, embedded AI tools offer a more cohesive user experience. Users can access AI assistance without needing to switch between applications, thereby minimizing friction and increasing engagement (Barrett et al., 2016). This distinction sheds new light on how different modes of utilizing generative AI—external vs. embedded—can lead to contrasting outcomes, offering a complementary understanding of AI's role in user behavior. In conclusion, while external generative AI tools like ChatGPT provide powerful capabilities for knowledge generation and assistance, embedded generative AI offers distinct advantages in the context of knowledge-sharing platforms.

Additionally, our research advances the understanding of human-AI interaction in the field of knowledge sharing. While prior literature hints that human users may either imitate or differentiate their responses from generative AI (e.g., Lysyakov & Viswanathan, 2023; Shan & Qiu, 2023), we are the first to systematically theorize and empirically test the framework of AI conformity and differentiation. Combining a natural experiment with a randomized experiment, we provide consistent evidence that the AI-conformity effect explains the increase in human contributor participation following the introduction of generative AI answers. In doing so, our research adds to the literature on human conformity to machines (e.g., Köbis et al., 2021; Logg et al., 2019; Salomons et al., 2021) and highlights the important role of informational influence in shaping knowledge-sharing activities in the presence of AI.

Furthermore, we examine how the quality of generative AI answers interacts with the introduction of generative AI answers to influence the content of subsequent human answers.

Details of these additional analyses are provided in Appendix F. These results are consistent with our explanation of the AI-conformity effect through the informational influence of generative AI answers, suggesting that when generative AI answers are of higher quality, they serve as more effective information sources, thereby encouraging greater conformity to AI content.

From a managerial standpoint, our research provides valuable insights for knowledge-sharing communities. While prior studies predominantly focus on the influence of external changes in AI technology, such as the launch of ChatGPT (e.g., Burtch et al., 2024; Xue et al., 2023), our findings shed light on how a platform-level policy, which platforms can directly control (Wang & Zheng, 2024), can alter knowledge-sharing dynamics. By examining the policy implemented by Quora, we find that the introduction of generative AI answers leads to an increase in contributor participation. However, we also highlight a potential downside: AI-generated answers may dampen the diversity of perspectives. Therefore, for managers of knowledge-sharing communities, our findings suggest potential benefits of incorporating generative AI into these platforms while also emphasizing the importance of addressing possible drawbacks that should not be ignored.

Moreover, our findings indicate that the effects of generative AI answers vary across different types of questions. Recognizing that generative AI answers have different effects on objective versus subjective questions, as well as on under-answered versus well-answered questions, platform managers may consider selectively deploying these AI answers to optimize user knowledge contributions. For instance, for questions that are more objective or under-answered, platforms may anticipate a greater increase in contributor participation following the introduction of generative AI answers. Additionally, we demonstrate that the effects vary across different topics. Details of this additional analysis are provided in Appendix G. The findings on the heterogeneity of effects can provide managers with valuable insights into optimizing the benefits of integrating AI tools within their communities.

Furthermore, our research pays particular attention to the changes in contributions from experts and non-experts following the introduction of generative AI answers. Our findings carry significant managerial implications. The marked increase in participation, particularly from non-expert contributors, highlights the potential of AI-generated answers to lower entry barriers for less experienced users by providing a reliable information source. This suggests that platform managers can leverage AI-generated content as a strategic tool to enhance non-expert participation in knowledge-sharing activities, potentially expanding the active user base and fostering greater inclusivity. However, it is essential for managers to strike a careful balance when integrating AI-generated answers. While sustaining high-quality contributions from experts remains critical to maintaining the platform's overall credibility and content quality, encouraging non-expert participation is valuable for platform growth and engagement. Therefore, platform managers should design AI-supported mechanisms that both preserve the active involvement of experts and motivate non-experts to contribute. By maintaining this balance, platforms can benefit from increased user engagement across experience levels while safeguarding the integrity and richness of the knowledge shared (Hou & Ma, 2022; Jin et al., 2021).

6.2 Limitations and Future Research

We acknowledge certain limitations in our research. One limitation of our study is that our findings rely on data obtained from a single Q&A platform, Quora, and its particular implementation of generative AI. However, it is worth noting that both the Quora platform and the generative AI tool used in our research are considered leaders in their respective fields. Additionally, while Q&A platforms represent just one format of the broader landscape of knowledge-sharing activities, the significance and widespread use of Q&A platforms reinforce the value of our results. Also, we recognize the constraints posed by the availability of data in our study. Future research can analyze metrics such as the time spent on crafting answers when such data becomes available.

In our research context, Quora has not publicly disclosed the criteria for selecting questions to receive generative AI answers. Therefore, we cannot confirm how questions are chosen for generative AI answers. Nevertheless, in our dataset, we observe pairs of nearly identical questions, where one question in a pair has a generative AI answer and the other does not. We have conducted a robustness check using a subsample of content-matched questions in Section 4.2.3. The results from this subsample are consistent with our main results using the full-sample dataset. Therefore, we believe that the potential selection bias does not significantly distort our findings.

Our study also opens multiple avenues for future explorations. Firstly, while our investigation primarily centers on the perspective of knowledge contributors, future studies could complement this research by considering the perspective of knowledge seekers. This would offer a more holistic understanding of how generative AI shapes the dynamics of knowledge sharing. Moreover, with the rapid evolution of generative AI technology, researchers should closely monitor and investigate the advancements of generative AI and its effects on knowledge sharing. For instance, emerging models capable of processing other modes of content such as images (e.g., GPT-4V; OpenAI, 2023b) and videos (e.g., Sora; OpenAI, 2024) may alter the effects of generative AI on knowledge-sharing dynamics by enabling the analysis and production of multimodal contents.

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APPENDICES

Appendix A

Table A.1. Comparison of Related Work and Our Study

Reference	Title	Focal policy (Policy nature)	Generative AI tool nature	Users' autonomy to adopt	Focal platform (Platform nature)
Huang et al. (2023)	Generative AI and content-creator economy: Evidence from online content creation platforms	Lofter: adoption of in-platform generative AI tool (Internal); Graffiti Kingdom: ban the use of generative AI (Internal)	Lofter: Embedded Graffiti Kingdom: External	Active	Lofter; Graffiti Kingdom (Social media)
Zhang et al. (2024)	How does AI-generated voice affect online video creation? Evidence from TikTok	TikTok adoption of AI-generated voice tool (Internal)	Embedded	Active	TikTok (Social media)
Knight et al. (2024)	Generative AI and user-generated content: Evidence from online reviews	ChatGPT launch (External)	External	Active	Yelp.com; Amazon.com (E-commerce)
Burtch et al. (2024)	The consequences of generative AI for UGC and online community engagement	ChatGPT launch (External)	External	Active	Stack Overflow; Reddit (Knowledge-sharing)
Li and Kim (2024)	Unveiling the effects of LLMs: Shifting UGC contribution in an online coding community	ChatGPT launch (External)	External	Active	Stack Overflow (Knowledge-sharing)
Sanatizadeh et al. (2023)	Information foraging in the era of AI: Exploring the effect of ChatGPT on digital Q&A platforms	ChatGPT launch (External)	External	Active	Stack Exchange (Knowledge-sharing)
Xue et al. (2023)	Can ChatGPT kill user-generated Q&A platforms?	ChatGPT launch (External)	External	Active	Stack Overflow (Knowledge-sharing)
Wang and Zheng (2024)	Can banning AI-generated content save user-generated Q&A platforms?	StackExchange ban of creating answers with generative AI (Internal)	External	Active	Stack Exchange (Knowledge-sharing)
Our study	**Generative AI and human knowledge sharing: Evidence from a natural experiment**	**Quora adoption of AI-generated answers (Internal)**	**Embedded**	**Passive**	**Quora (Knowledge-sharing)**

Our study differentiates itself from existing work in the following ways:

(i) **Embedded vs. external generative AI tools**: Our study diverges from previous work by focusing on the context of embedded AI tools (generative AI systems that are integrated directly into existing platforms, rather than functioning as standalone tools), which operate differently from external tools like ChatGPT. Specifically, while prior studies suggest that external tools act as substitutes and reduce platform engagement (i.e., Burtch et al., 2024; Li & Kim, 2024; Sanatizadeh et al., 2023; Xue et al., 2023), our findings indicate that embedded generative AI can potentially increase user participation by providing users with relevant information in a seamless manner. This integration allows the AI to operate within the context of specific tasks, enhancing engagement and user contributions.

(ii) **Reactive vs. active adoption of generative AI**: Unlike previous studies focusing on active adoption, where users deliberately seek out AI-generated content (e.g., Burtch et al., 2024; Sanatizadeh et al., 2023), our research explores how users respond when AI-generated answers are presented passively, without actively searching for them. Our results reveal that this passive exposure provides contributors with a reliable source of conformity by making relevant information readily available, which in turn encourages users to engage in answering questions.

(iii) **Relational vs. standalone quality examination**: While previous studies (e.g., Wang & Zheng, 2024; Zhang et al., 2024) have primarily focused on the "standalone" quality of user-generated content, our research investigates a complementary aspect: the relationship between subsequent user-generated content and AI-generated content when generative AI is integrated into the platform. Our findings indicate that, following the introduction of AI-generated answers, user contributions tend to become similar to the AI-generated content. This suggests potential tradeoffs of incorporating AI-generated answers: Despite the increase in participation, there may be a reduction in the diversity of perspectives, limiting the richness and informativeness of the answers.

Appendix B

In our study, we rigorously comply with Quora's Terms of Service to ensure that all data usage and research practices align with the platform's legal and ethical guidelines. Specifically, according to Quora's Terms of Service (https://www.quora.com/about/tos), users are not allowed to access, search, or collect data from the Quora Platform "(1) to create derivative works of Our Content and Materials; (2) to train or develop any AI, large language models or machine learning algorithms on Our Content or Materials; (3) to create any service competitive to the Quora Platform; or (4) for other commercial purposes except as expressly permitted by these Terms of Service or the written consent of Quora." We collected a small portion of data exclusively for academic research purposes, ensuring that our data collection practices remain outside the scope of the restricted uses defined by the platform. Additionally, consistent with established methodologies in the literature (e.g., Das & Semaan, 2022; Wang et al., 2013), we implemented a rate limit of 10 requests per second during data collection to minimize any potential burden on Quora's infrastructure and to adhere to best practices in ethical data usage.

To collect a sample of questions for our analysis, we leverage a unique feature of Quora that logs user activities. Each activity, including posting questions or answers, is assigned a unique identifier that denotes its chronological occurrence. For example, the identifier "2487898700" corresponds to activity at 7:26:44 PM on October 19, 2022, and a subsequent number, "2487898815," identifies a closely following activity at 7:26:48 PM on the same day. This precision in logging enables us to accurately sample our data.

Appendix C

Instruction for GPT-4 to categorize questions as objective or subjective:

Classify the following question as "subjective" or "objective": [question content]. On a scale of 1 to 9, with 1 being "not at all confident" and 9 being "extremely confident", rate how confident you feel about the answer. Your answer should be just like “subjective/objective [confidence level]”.

Instruction for GPT-4 to categorize questions as open-ended or closed:

Classify the following question as “open-ended” or “closed”: [question content]. On a scale of 1 to 9, with 1 being “not at all confident” and 9 being “extremely confident”, rate how confident you feel about the answer. Your answer should be just like “open-ended/closed [confidence level]”.

Instruction for GPT-4 to measure human-AI answer similarity:

Given the question “[question content]”, from a semantic perspective, please assess the content similarity between the following two answers. Then rate how the second answer is similar to the first from 0% to 100%, with 0% being “totally different” and 100% being “exactly the same”. The first answer is: “[AI answer]”. The second answer is: “[user answer]”. Your response should be just a whole number percentage from 0% to 100%.

Appendix D

Table D.1. Summary Statistics of Content-Matched Question Subsample

Variables	Mean (Treated)	Mean (Control)	Difference
QuesTenure	55.146	50.677	4.469
AccNumAnswer	165.708	104.129	61.579
AccNumCollap	36.646	33.968	2.678
LastAnswerInterval	20.230	22.645	-2.415
QuesFollow	50.188	21.710	28.478
QuesLen	50.583	55.968	-5.385
QuesView	371089.6	191467.7	179621.9
QuesOpen	0.563	0.516	0.047
QuesFromTop	0.021	0.000	0.021
QuesSmog	7.265	6.650	0.615

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix E

Table E.1. Impact of Generative AI Answer: Moderating Role of Question Objectivity (only include questions that are GPT-4 is “extremely confident” to categorize)

Variables	(1) log(NumAnswer)	(2) log(AveLenAnswer)

AIAnswered	0.034^{**}	0.101^{*}
	(0.015)	(0.060)
AIAnswered × QuesObjective	0.053^{***}	0.246^{***}
	(0.016)	(0.063)
Control variables	Yes	Yes
Question fixed effects	Yes	Yes
Week fixed effects	Yes	Yes
Observations	98,825	98,825

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$; Robust standard errors clustered at the question level are in parentheses.

Appendix F

The Effect on Human-AI Answer Similarity: Moderating Role of AI Answer Quality

To further explore the conditions under which human contributors repeat generative AI answers more, we incorporate the quality of generative AI answers as a moderating factor. Specifically, we recruit experienced Q&A platform users from Prolific Academic[1] to serve as human raters, evaluating the overall quality of generative AI answers. Each rater evaluates 10 randomly selected AI-generated answers from a pool of 2,319 answers in our treatment group. For each answer, the rater receives both the question and the answer, without being informed that the answer is generated by AI. Raters indicate to what extent they think the answer is of high quality on a 9-point scale (1 = not at all, 9 = very) after reading both the question and the answer. Each answer is rated by at least two raters, and the final score for each answer is determined by averaging the ratings.

We use the content-matched question subsample for this analysis. For a question in the treatment group (where a generative AI answer is provided), the AI answer quality is based on the ratings from human raters mentioned earlier. For a question in the control group (where no AI answer is shown), the AI answer quality score is given as that of their counterpart question in the treatment group in its question set, or the average of multiple counterpart questions in the treatment group if the question set contains more than one. We proceed to generate a dummy variable, $AIAnsQuality$, which equals 1 if the AI-generated answer score is above the median and 0 otherwise. Afterward, we perform the following regressions:

[1] https://www.prolific.com/.

$$Z_{ijt} = \beta_0 + \beta_1 AIAnswered_{ijt} + \beta_2 AIAnswered_{ijt} \times AIAnsQuality_i + X_{ijt} + \xi_j + \delta_t + \varepsilon_{ijt} \quad (F.1)$$

where Z_{ijt} includes two measures of human-AI answer similarity, $AveSimGPT_{ijt}$ and $AveSimSBERT_{ijt}$. The results are presented in Table F.1. The positive coefficients of the interaction between $AIAnswered$ and $AIAnsQuality$ indicate that introducing generative AI answers has greater positive effects on human-AI answer similarity when the generative AI answers have higher quality. This finding indicates that high-quality AI answers encourage human contributors to incorporate more of the AI content, as these answers serve as a more effective information source for conformity. Consequently, contributors are more inclined to exploit the high-quality AI-generated answers and then contribute subsequently.

Table F.1. Moderating Effects of AI Answer Quality

Variables	(1) AveSimGPT	(2) AveSimSBERT
AIAnswered	0.030**	0.061**
	(0.014)	(0.026)
AIAnswered × AIAnsQuality	0.061**	0.070**
	(0.027)	(0.036)
Control variables	Yes	Yes
Set fixed effects	Yes	Yes
Week fixed effects	Yes	Yes
Observations	2,049	2,049

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; Robust standard errors clustered at the set level are in parentheses.

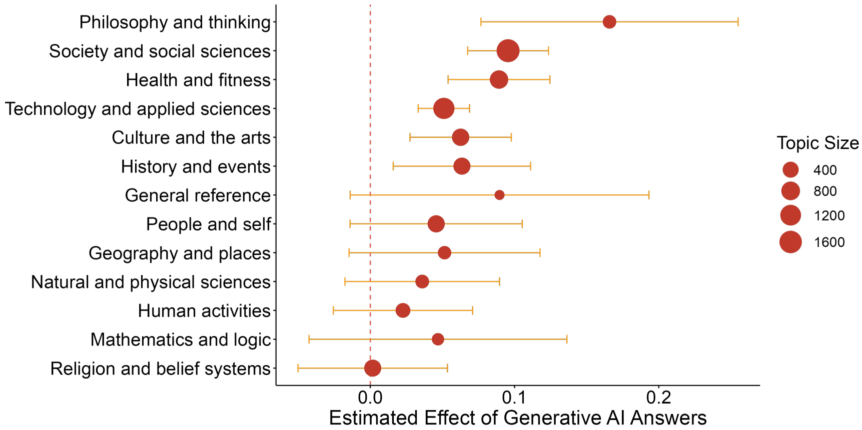
Appendix G

Heterogeneity of Effects Across Question Topics

We further explore question types by utilizing OpenAI's GPT-4 model to categorize questions into 13 standard topics based on Wikipedia classifications.[2] Following prior research (Burtch et al., 2024), we examine the heterogeneity in the effects of generative AI answers across these topics. Figure G.1. presents our results, showing that the effects of generative AI answers on the volume of subsequent human answers are positive across all topics. We can observe

[2] https://en.wikipedia.org/wiki/Wikipedia:Contents/Categories.

significant effects among the most popular topics on Quora, such as Technology and applied sciences, Society and social sciences, and Health and fitness.



Notes: Estimates are derived from a DID regression conducted for each topic. The figure illustrates the effect estimates for topics, accompanied by 95% confidence intervals. The topic size refers to the number of questions for each topic.

Figure G.1. Topic-specific effects of Generative AI answers on Quora

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